Palindromic Substring

s = abc

substr ⇒ "a", "b", "c", "ab", "ac", "bc", "abc"

Hunny → H

Substring ⇒ Hu, Hun, Hunn, Hunny } 5 + un, unn, uny } 9 + nn, ny } 12 + ny + y } = 15

abc ⇒ substr ⇒ "a", "b", "c", "ab", "ac", "bc", "abc"

palindrome ⇒ a, b, c only.

brute force ⇒ find Substring $O(n^2)$

{ validPalindrome() $O(n)$

}

→ $O(n^3)$

$O(n^2)$ ⇒

0	1	2
n	o	o	n

i j

odd

① palindrome ⇒ Yes (s[i] == s[j]) ①

② | n | o | o | n |

i j

palindrome ⇒ Yes (s[i] == s[j]) ②

| n | o | o | n |

i j

noo → palindrome ⇒ No ! stop

③ | n | o | o | n |

i j

o → palindrome → Yes! ③

| n | o | o | n |

i j

oon → palindrome ⇒ NO! ! Stop

④ | n | o | o | n |

i j

n → palindrome ⇒ Yes! ④

| n | o | o | n |

i j → stop

n

substring → Odd, even

even

| n | o | o | n |

i j

① no → palindrome ⇒ NO!

② | n | o | o | n |

i j

oo → palindrome → Yes! ①

| n | o | o | n |

i j

noon → palindrome → Yes! ②

| n | o | o | n |

i j

→ ! stop

③ | n | o | o | n |

i j

on → palindrome ⇒ No!

| n | o | o | n |

i j

→ ! Stop.

Odd ⇒ 4

even ⇒ 2 = ⑥

⇒ noon

n, o, o, n, no, noo, noon, oo, oon, on

6 palindrome

logic ⇒

0	1	2	3	4	5
b	a	b	b	a	r

i

odd

b – 1

bba – 3

abbar – 5 } odd

even

0	1	2	3	4	5
b	a	b	b	a	r

i j

bb – 2

abba – 4

babbar – 6 } even